Beacon Summary Chart

Sixdof Space
ULTRA FAST OPTICAL TRACKING

Beacon	Part #	Power Level	Min Distance	Max Distance*	Field of View**
Low Power LED Beacons	030-1001-000, 030-1001-001, 030-1001-002, 030-1001-003		0.05m	8m	160°
HYF-N LED without Lens	030-1006-000, 030-1006-002				130°
		2A	0.2m	12.1m	
		4A	0.23m	15.3m	
		6A	0.25m	19.5m	
HYF-N LED with Lens	030-1006-001, 030-1006-003				90°
		2A	0.3m	16.7m	
		4A	0.33m	23.6m	
		6A	0.35m	28.4m	
HYF-S LED without Lens	030-1011-000, 030-1011-002				130°
		2A	0.25m	14.3m	
		4A	0.27m	19.5m	
		6A	0.35m	23.5m	
HYF-S LED with Lens	030-1011-001, 030-1011-003				90°
		2A	0.25m	19.9m	
		4A	0.44m	27.4m	
		6A	0.5m	37.7m	
High Power 100W LED	030-1016-000, 030-1016-001, 030-1016-002, 030-1016-003				130°
		2A	0.4m	16.3m	
		4A	0.45m	21.1m	
		6A	0.5m	26.3m	
High Power 100W with Cone	030-1016-004				70°
		2A	1.5m	24.9m	
		4A	1.5m	31.8m	
		6A	2.2m	40.5m	
HYF-120S-4 LED with Lens	030-1021-000, 030-1021-001		1.7m	62m	90°
VCSEL (Laser)	030-1026-000, 030-1026-001		2.7m	100m	20°
HYF-120S-16 LED with Lens, Integrated Beacon with Driver	020-1016-001		6.0m	100m	90°

*Maximum Distance shown assumes ideal alignment between beacon and sensor

**Defined as the 60% peak-intensity level

Revised Aug 9th, 2026

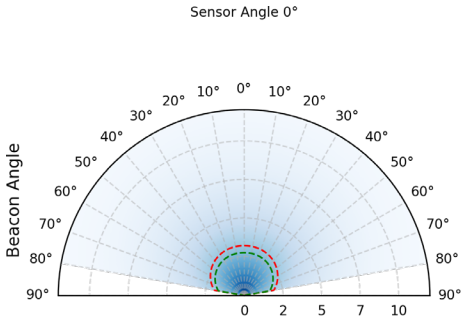
Low Power LED Beacon

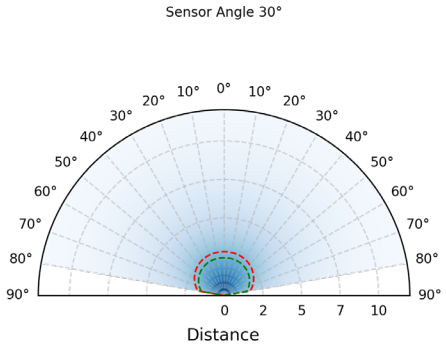


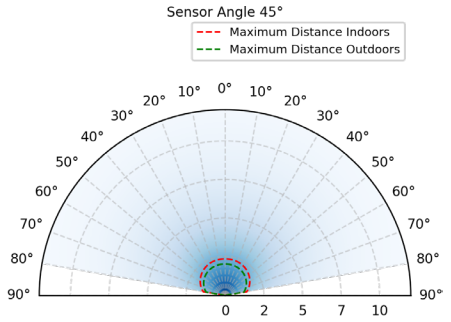
Max Angle	±80°
Dimensions	2.5 x 2.0 x 0.7cm

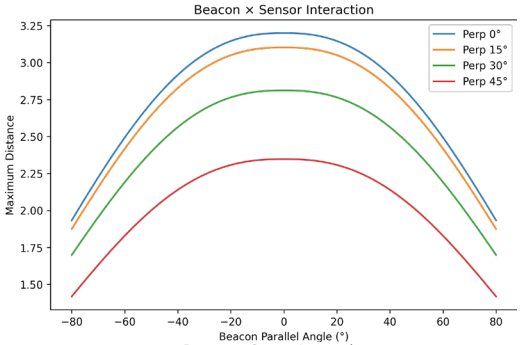
Part #	
030-1001-000	Low Power LED Beacon
030-1001-001	LED with 2Pin Connector
030-1001-002	LED with Housing, 1m Cable, Connector for Basalt
030-1001-003	LED with Housing, 1m Cable, Connector for Shale

Min Distance	0.05m
Max Distance	3.2m
Driver	Basalt-L
Power	0.3W

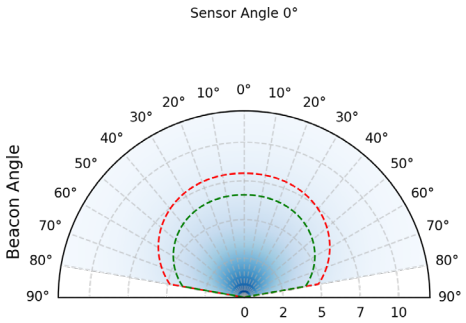


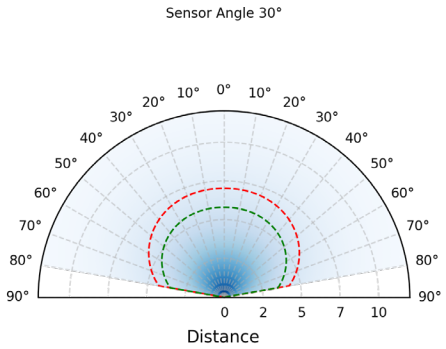


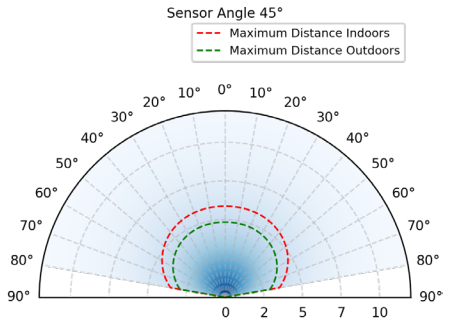


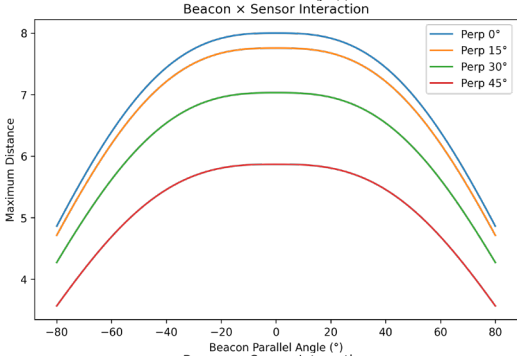


Min Distance	0.15m
Max Distance	8.0m
Driver	Basalt-D
Power	0.7W

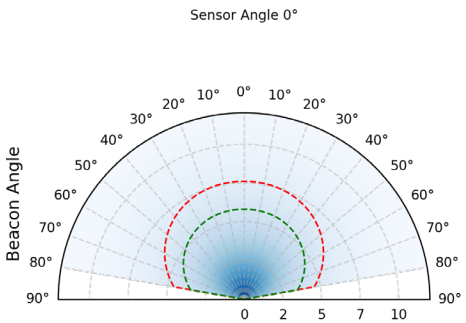


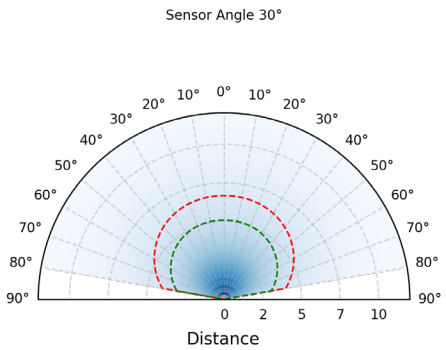


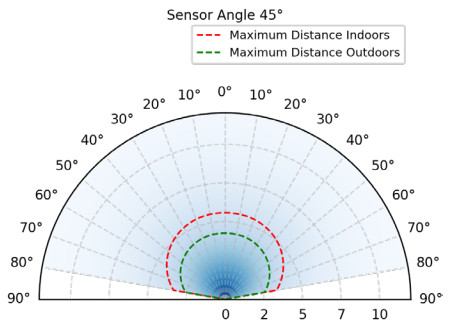


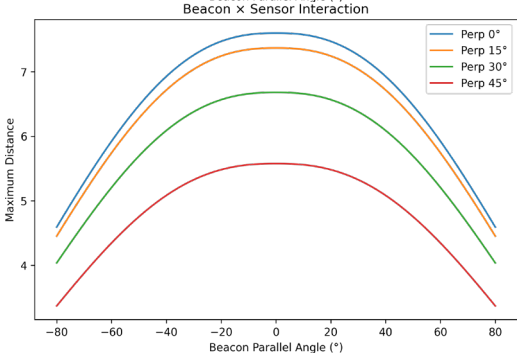


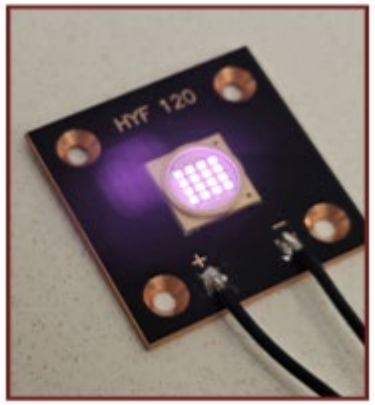
Min Distance	0.11m
Max Distance	7.6m
Driver	2A Shale
Power	0.5W











HYF-120N LED without Lens

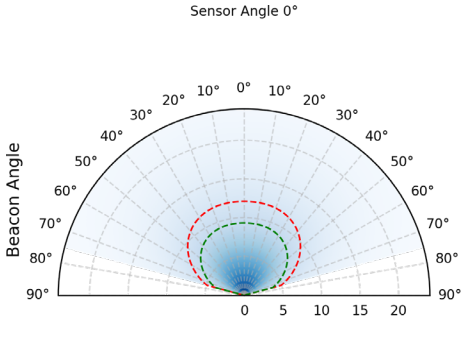
Sixdof Space
ULTRA FAST OPTICAL TRACKING

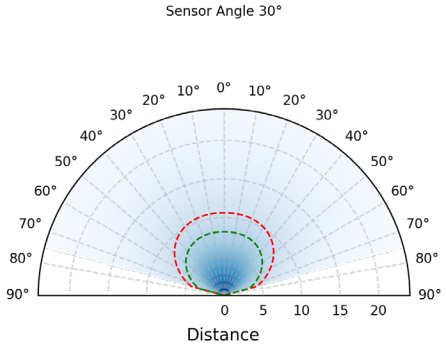
Max Angle	±65°
Dimensions	3.5 x 3.5cm

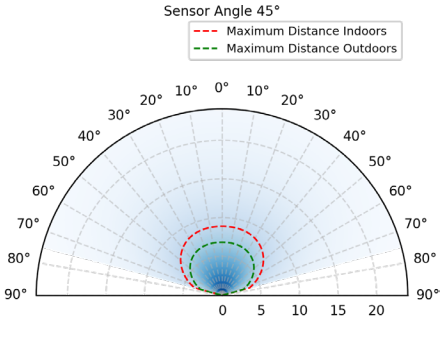
Part #

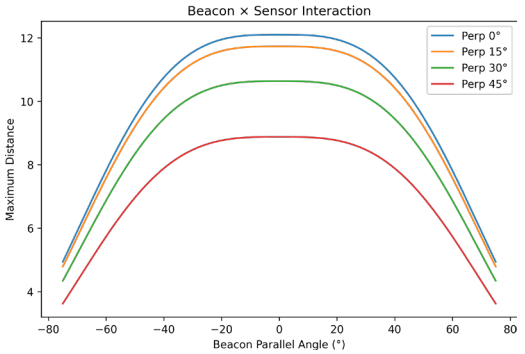
030-1006-000	HYF-120N without Lens
030-1006-002	HYF-120N without Lens, 1m Cable, Connector for Shale

Min Distance	0.20m
Max Distance	12.1m
Driver	Shale 2A
Power	0.8W

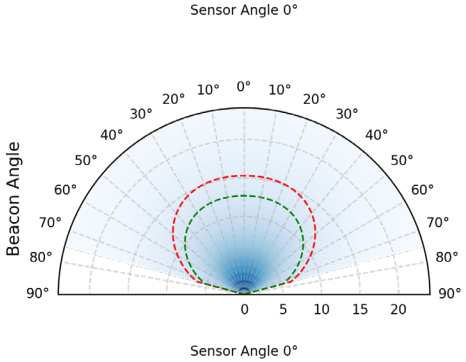


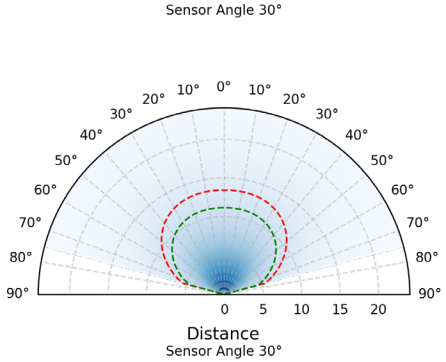


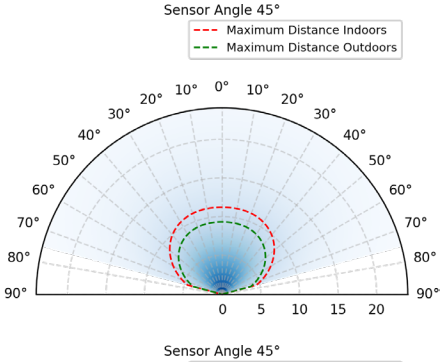


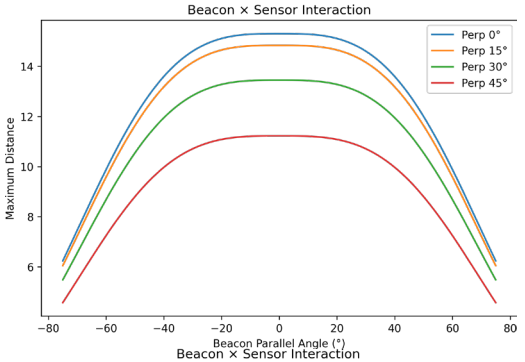


Min Distance	0.23m
Max Distance	15.3m
Driver	Shale 4A
Power	1.9W

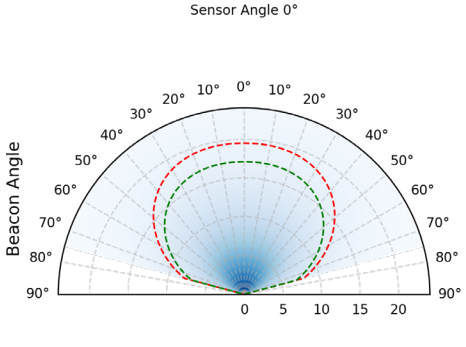


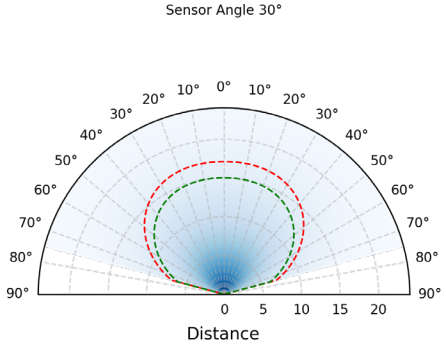


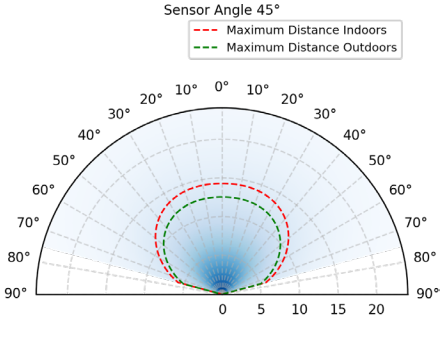


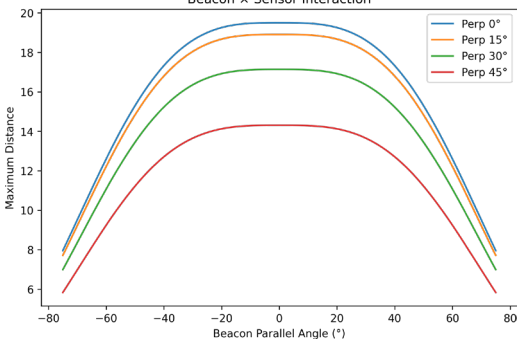


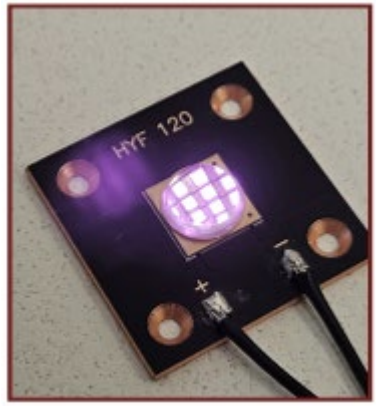
Min Distance	0.25m
Max Distance	19.5m
Driver	Shale 6A
Power	2.8W











HYF-120N LED with Lens

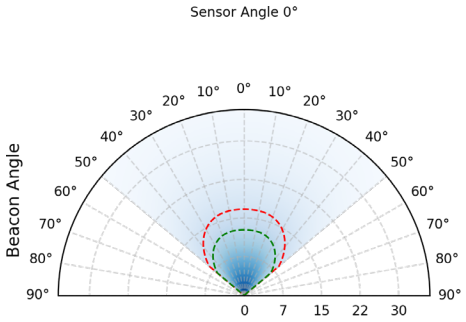
Sixdof Space
ULTRA FAST OPTICAL TRACKING

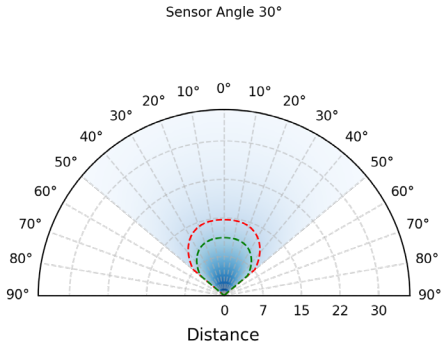
Max Angle	±45°
Dimensions	3.5 x 3.5cm

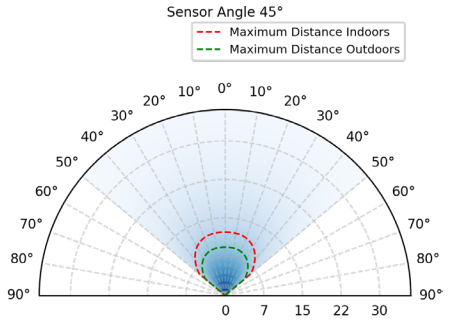
Part #

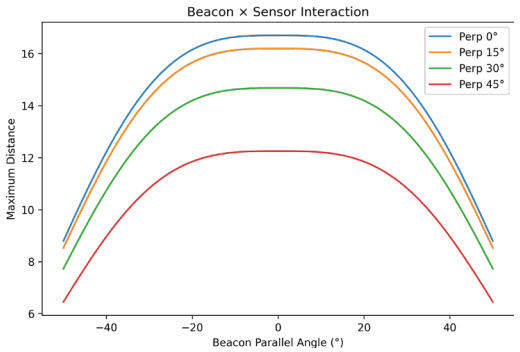
030-1006-001	HYF-120N with Lens
030-1006-003	HYF-120N with Lens, 1m Cable, Connector for Shale

Min Distance	0.30m
Max Distance	16.7m
Driver	Shale 2A
Power	0.8W

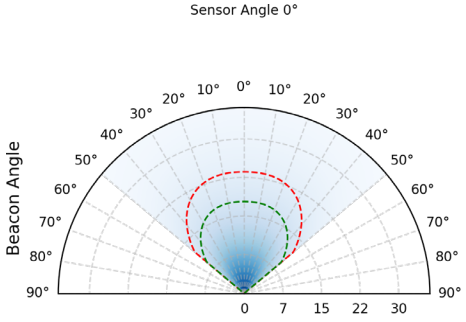


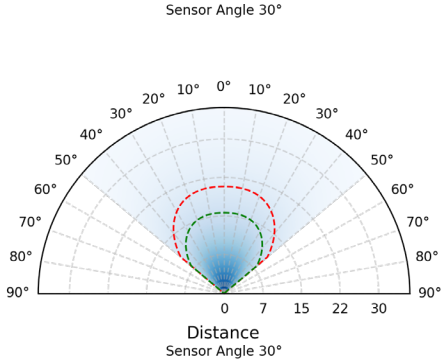


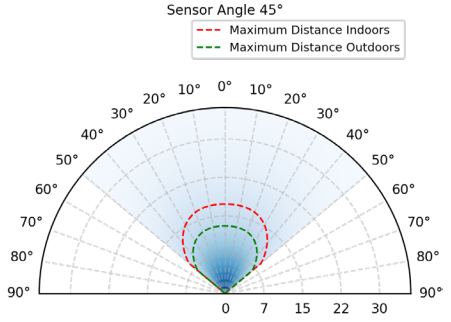


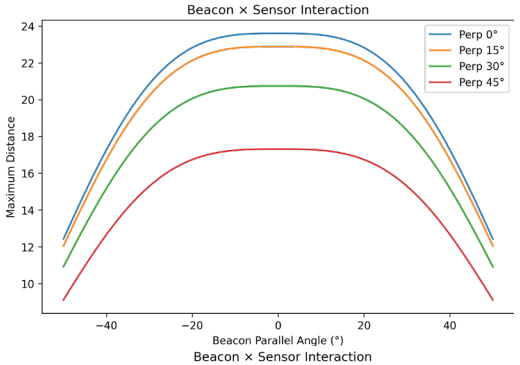


Min Distance	0.33m
Max Distance	23.6m
Driver	Shale 4A
Power	1.9W

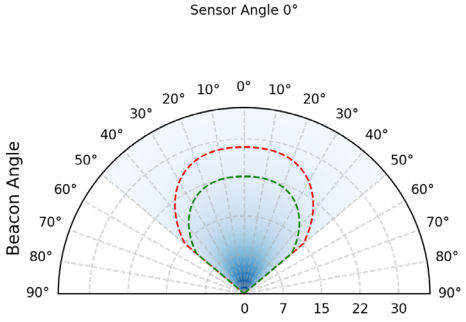


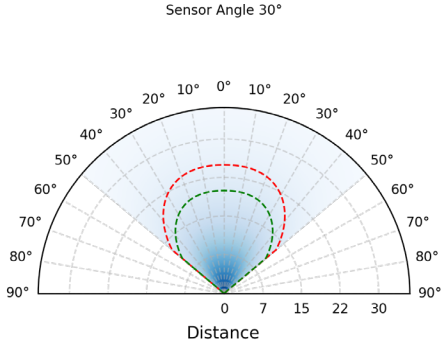


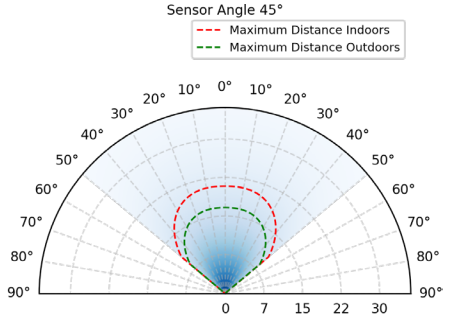


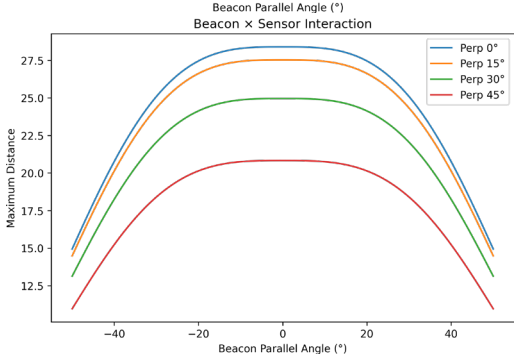


Min Distance	0.35m
Max Distance	28.4m
Driver	Shale 6A
Power	2.8W

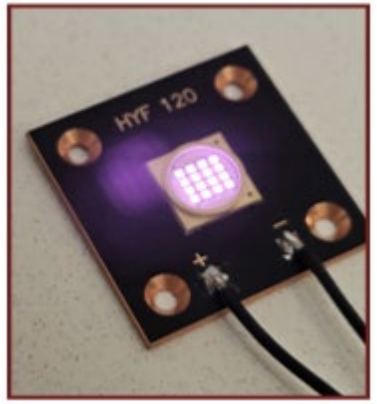








HYF-120S LED without Lens

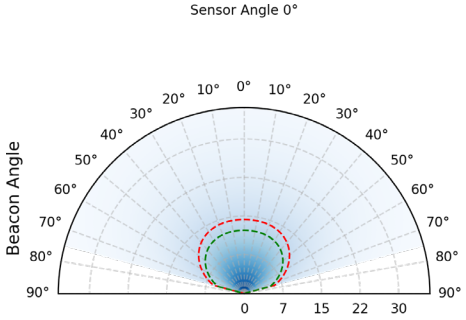


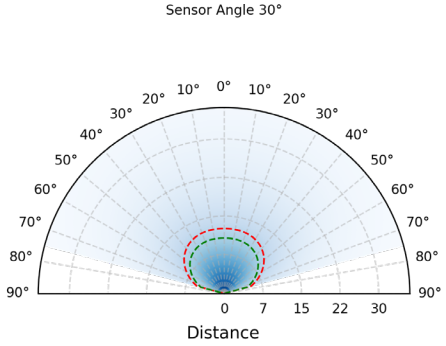
Max Angle	±65°
Dimensions	3.5 x 3.5cm

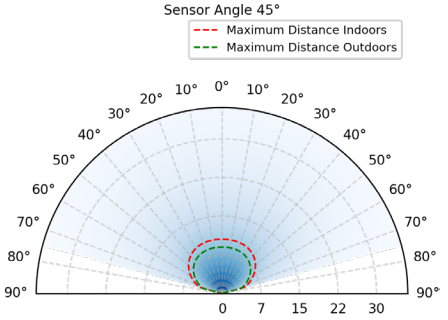
Part #

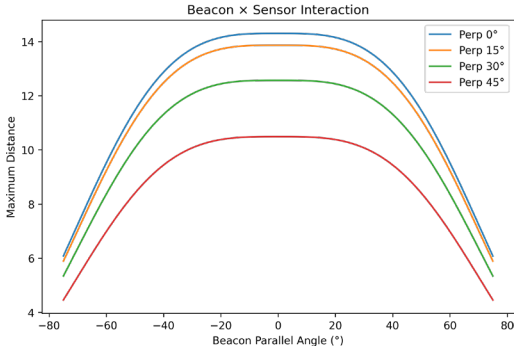
030-1011-000	HYF-120S without Lens
030-1011-002	HYF-120S without Lens, 1m Cable, Connector for Shale

Min Distance	0.25m
Max Distance	14.3m
Driver	Shale 2A
Power	1.3W

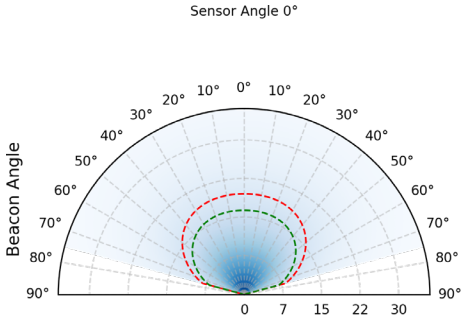


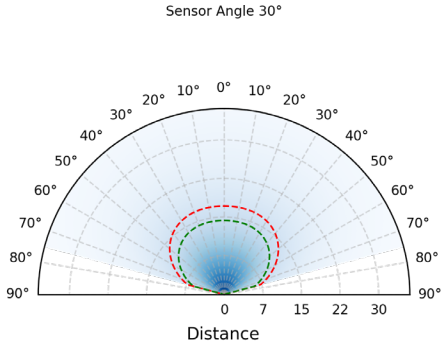


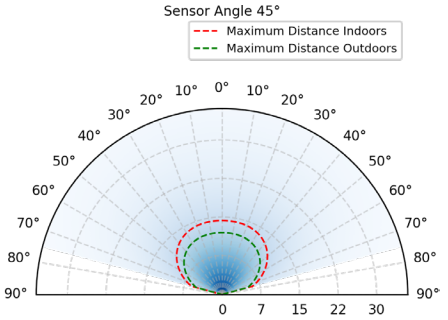


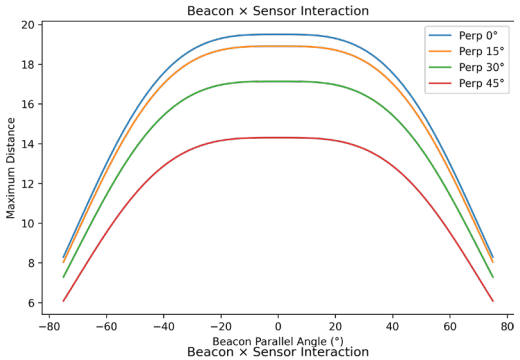


Min Distance	0.27m
Max Distance	19.5m
Driver	Shale 4A
Power	3W

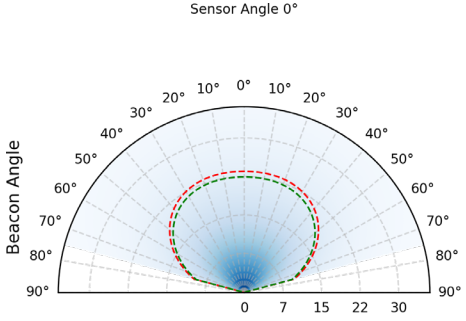


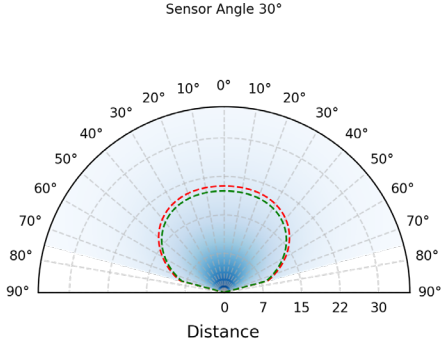


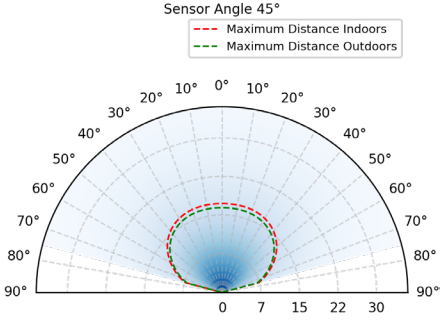


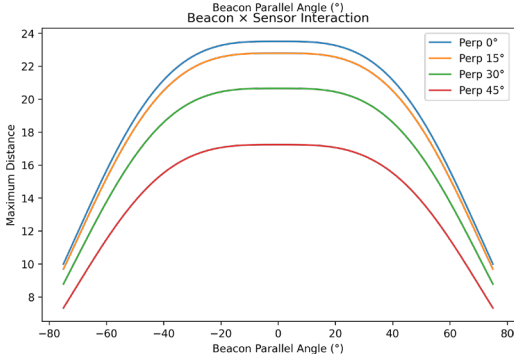


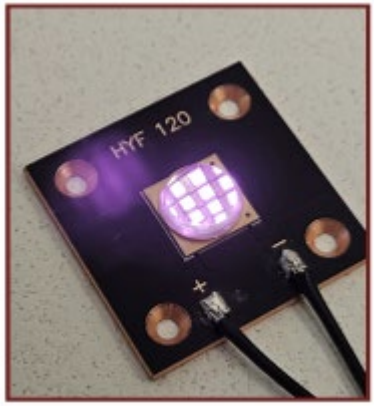
Min Distance	0.35m
Max Distance	23.5m
Driver	Shale 6A
Power	4.6W











LED HYF-120S with Lens

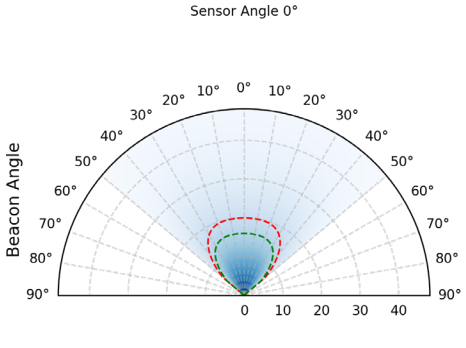
Sixdof Space
ULTRA FAST OPTICAL TRACKING

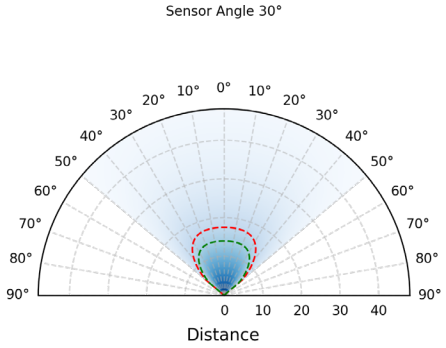
Max Angle	±45°
Dimensions	3.5 x 3.5cm

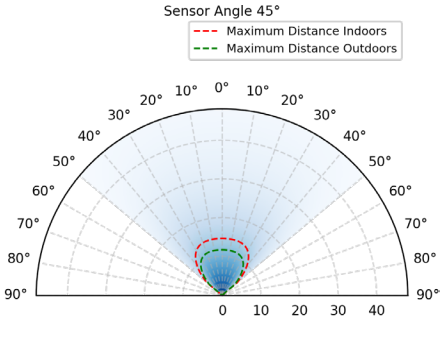
Part #

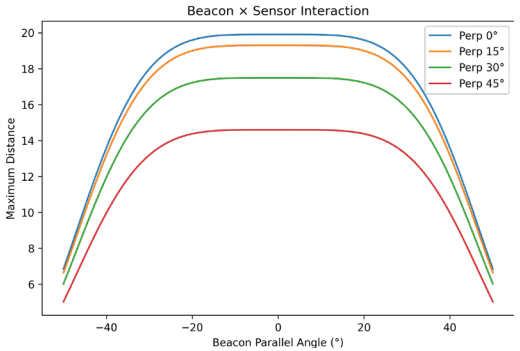
030-1011-001	HYF-120S with Lens
030-1011-003	HYF-120S with Lens, 1m Cable, Connector for Shale

Min Distance	0.25m
Max Distance	19.9m
Driver	Shale 2A
Power	1.3W

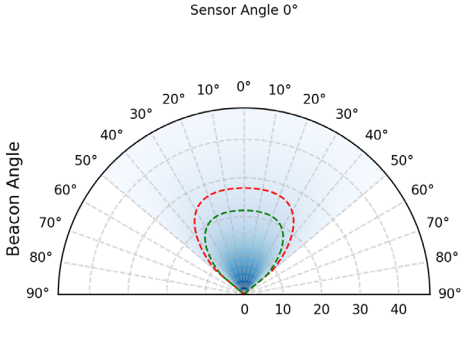


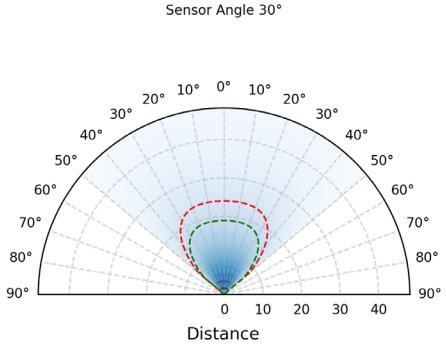


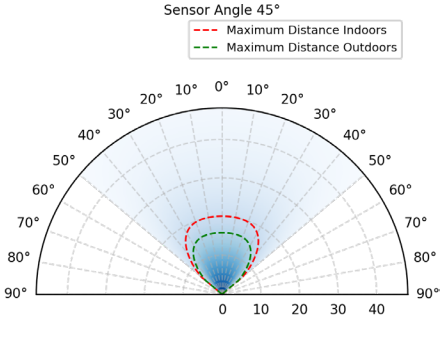


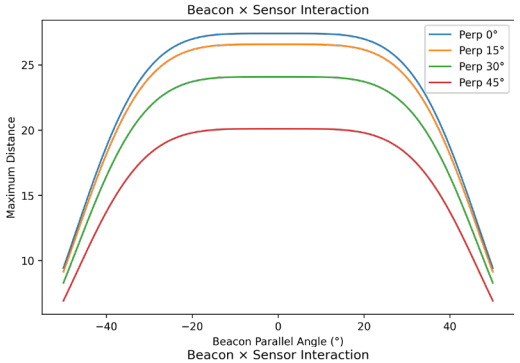


Min Distance	0.44m
Max Distance	27.4m
Driver	Shale 4A
Power	3W

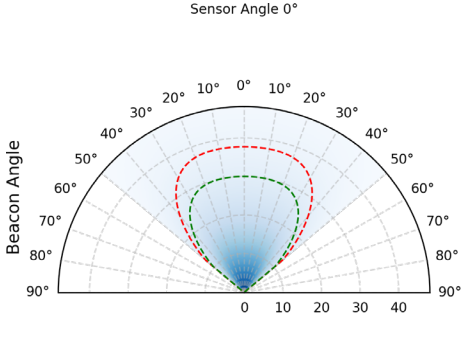


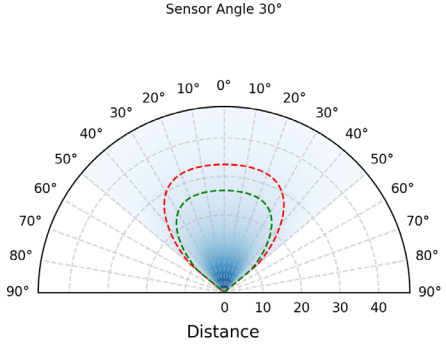


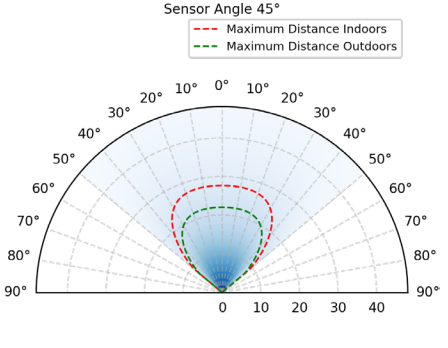


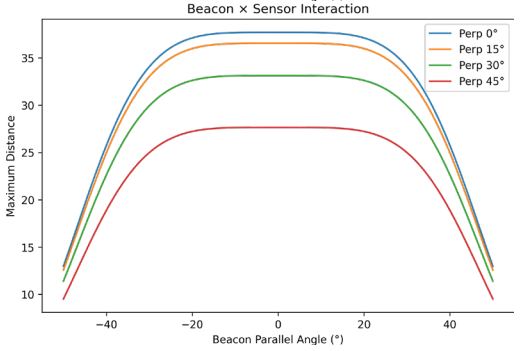


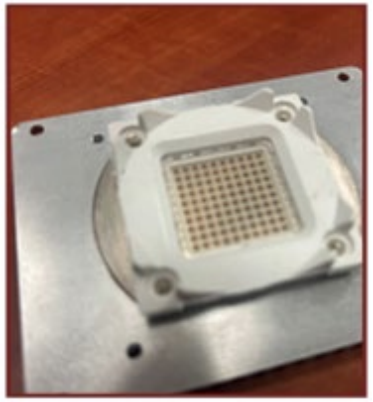
Min Distance	0.50m
Max Distance	37.7m
Driver	Shale 6A
Power	4.6W









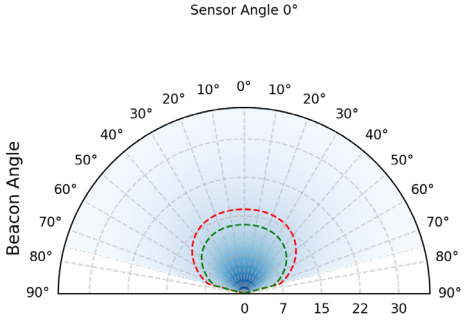


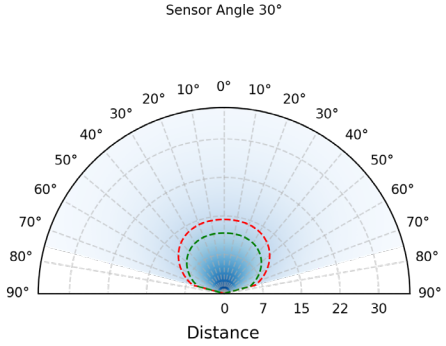
High Power 100W LED

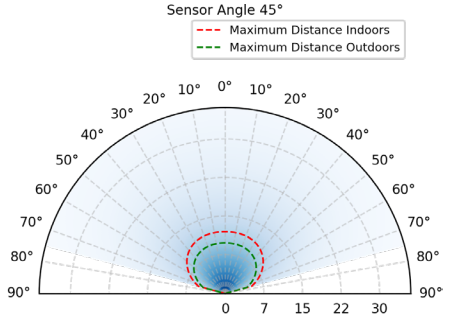
Max Angle	±65°
Dimensions	4.0 x 5.5cm

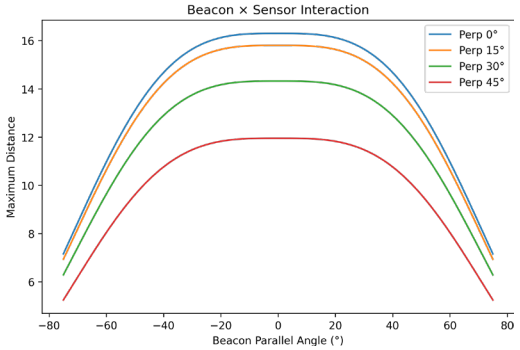
Part #	
030-1016-000	High Power 100W LED
030-1016-001	High Power 100W LED (with Heatsink)
030-1016-002	High Power 100W LED (with Heatsink, 1m Cable, Driver for Basalt)
030-1016-003	High Power 100W LED (with Heatsink, 1m Cable, Connector for Shale)

Min Distance	0.40m
Max Distance	16.3m
Driver	Shale 2A
Power	1.5W

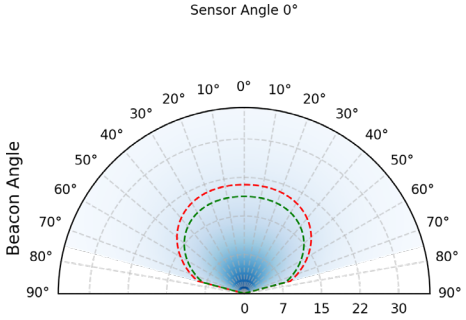


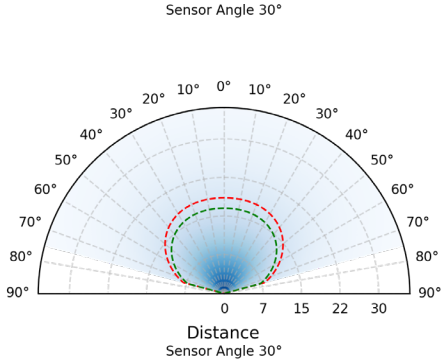


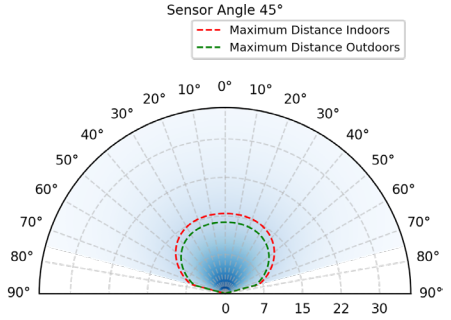


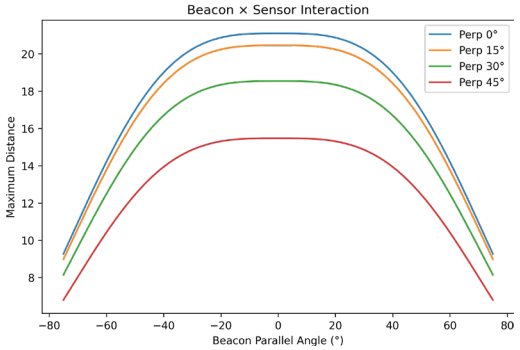


Min Distance	0.45m
Max Distance	21.1m
Driver	Shale 4A
Power	3.5W

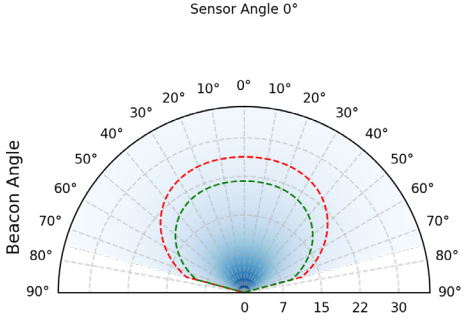


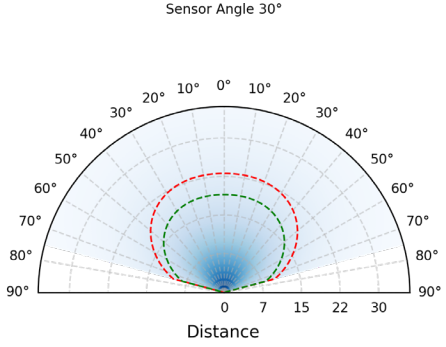


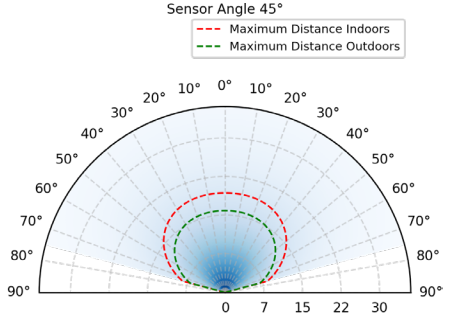


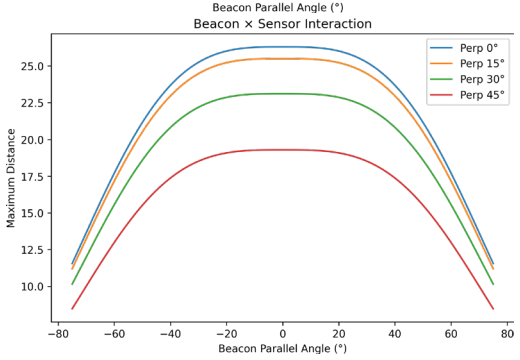


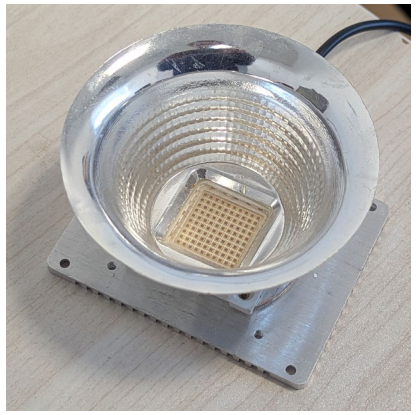
Min Distance	0.50m
Max Distance	26.3m
Driver	Shale 6A
Power	5.5W











High Power 100W LED with Cone

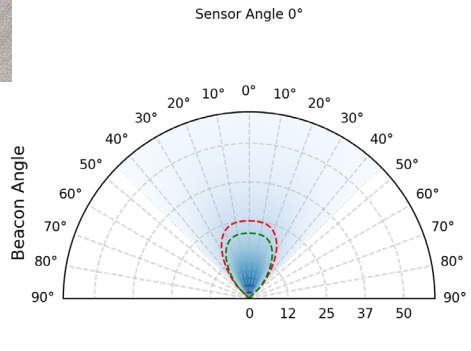
Sixdof Space
ULTRA FAST OPTICAL TRACKING

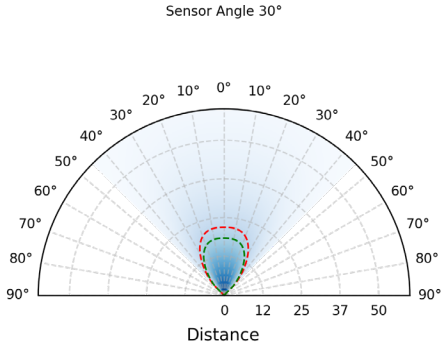
Part #
030-1016-004 High Power 100W LED with Cone (with Heatsink, 1m Cable, Connector for Shale)

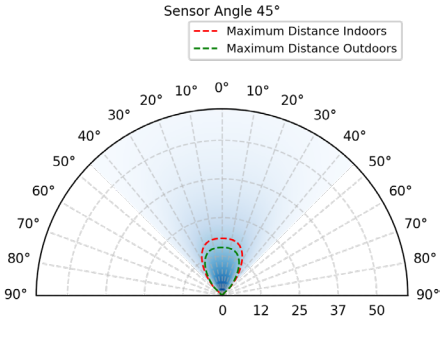
Max Angle ±35°

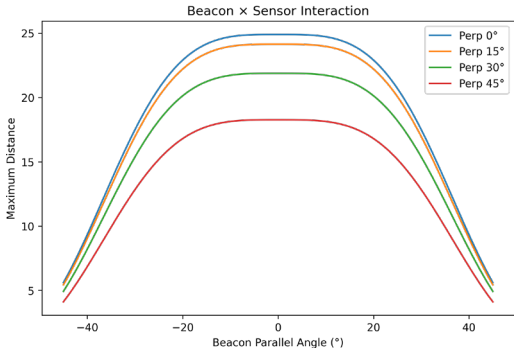
Dimensions 4.0 x 5.5 x 4.0cm

Min Distance	1.50m
Max Distance	24.9m
Driver	Shale 2A
Power	1.5W

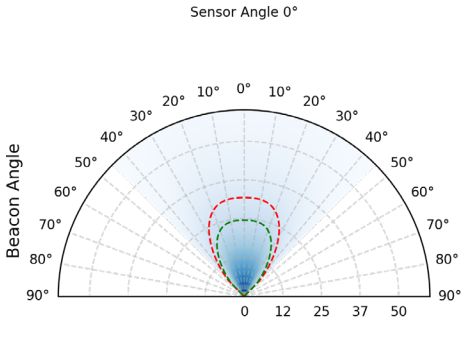


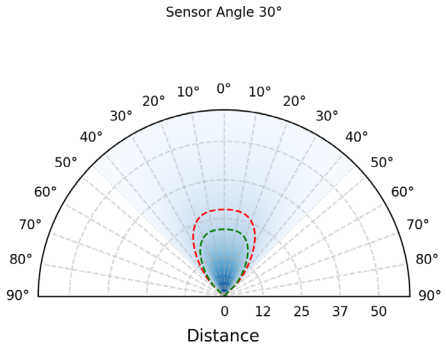


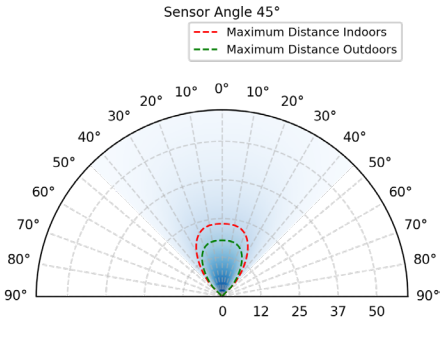


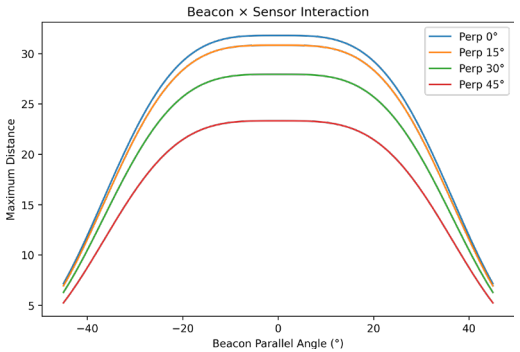


Min Distance	1.50m
Max Distance	31.8m
Driver	Shale 4A
Power	3.5W

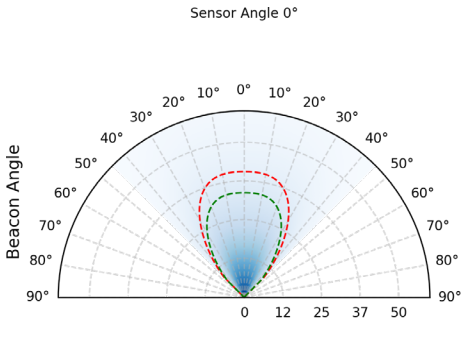


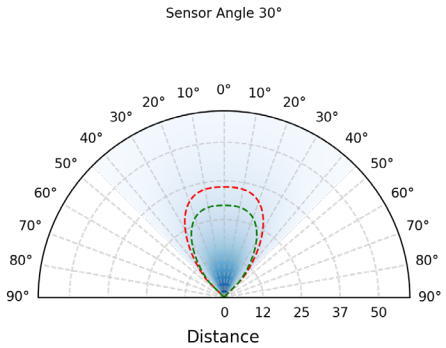


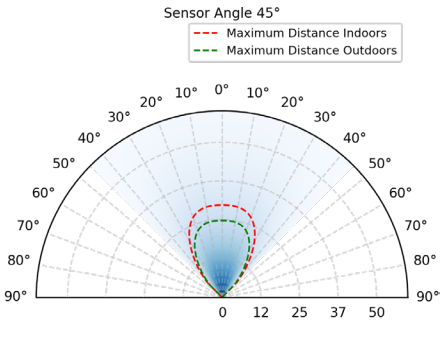


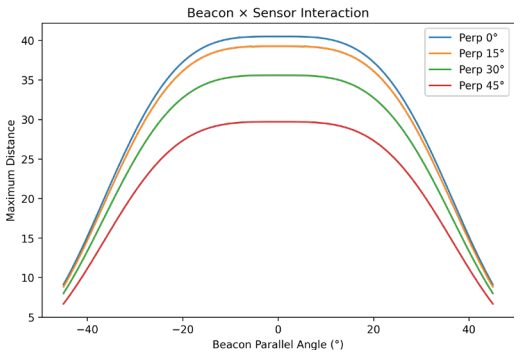


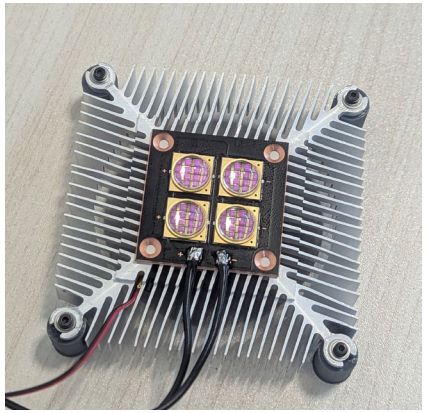
Min Distance	2.20m
Max Distance	40.5m
Driver	Shale 6A
Power	5.5W











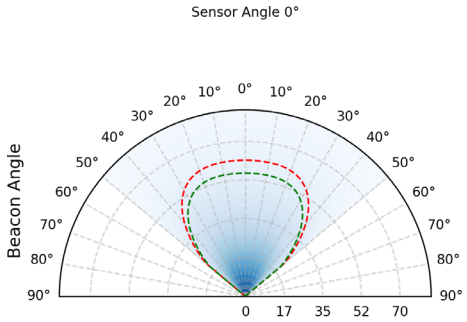
HYF-120S-4 LED with Lens

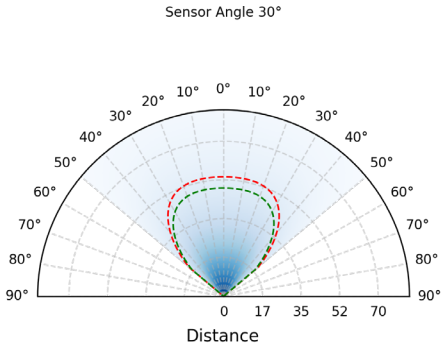
Max Angle	±45°
Dimensions	8.5 x 8.5 x 2.0cm

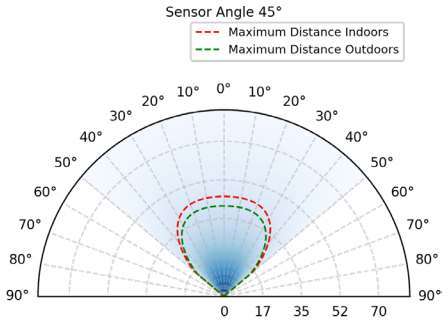
Part #

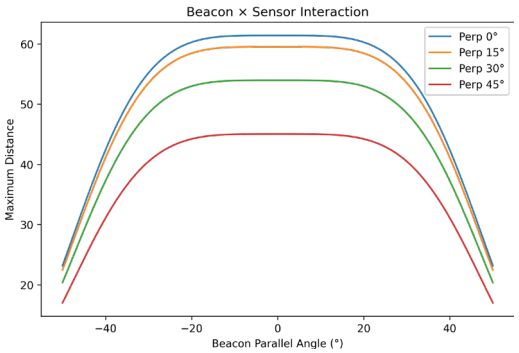
030-1021-000	HYF-120S-4 LED with Lens
030-1021-001	HYF-120S-4 LED with Lens (with 1m Cable, Connector for Zircon)

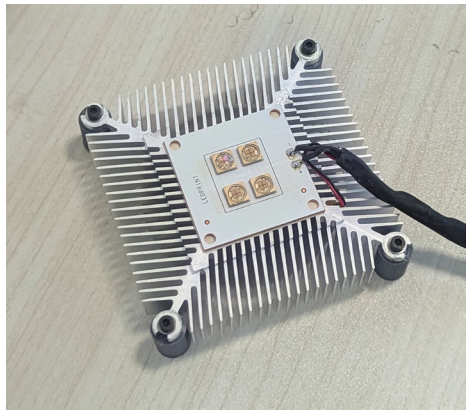
Min Distance	1.70m
Max Distance	61.4m
Driver	Zircon 24A
Power	19.5W











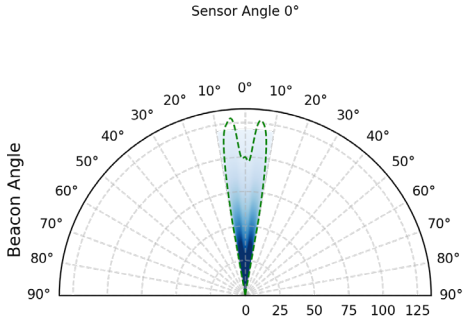
VCSEL

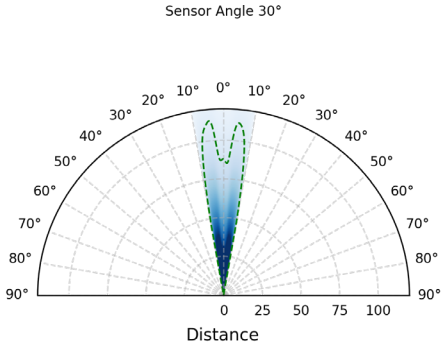
Max Angle	±10°
Dimensions	8.5 x 8.5 x 2.0cm

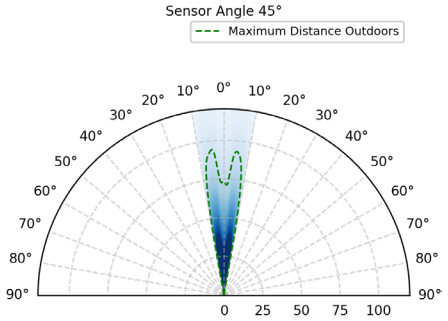
Part #

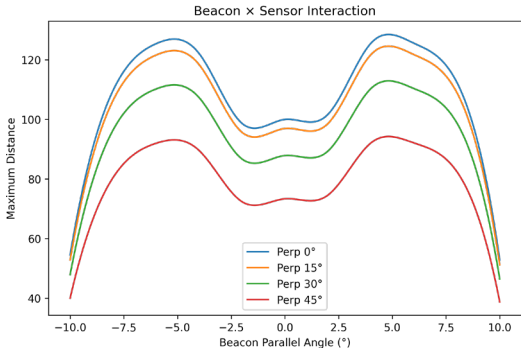
030-1026-000	VCSEL
030-1026-001	VCSEL (1m Cable, Connector for Zircon)

Min Distance	2.70m
Max Distance	100m
Driver	Zircon 20A
Power	15W









Eye Safety Warning:
Minimum safe viewing distance = 1.76 meters
We recommend eye safety glasses with OD 3 or higher for 850nm



HYFS16 100M LED Integrated Beacon with Driver

Sixdof Space
ULTRA FAST OPTICAL TRACKING

Max Angle	±45°
Dimensions	17 x 10 x 10cm
Power Requirements	110V/220V 50/60 Hz AC

Part #
020-1016-001

Min Distance	6m
Max Distance	100m
Power	144W

